

MAXWELL HOWE

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SUMMARY

Web GIS developer and spatial analyst (M.S. Geo-Information Science) who builds production mapping applications and rigorous spatial analyses end to end — from data acquisition, PostGIS database design, and statistical modeling through interactive front ends people actually use. Portfolio of live deployed applications spanning LiDAR terrain analysis, real-time transit feeds, statewide atlases, and reproducible R pipelines. Licensed mathematics educator with a proven ability to explain technical work to any audience. Seeking GIS developer, web GIS, and spatial analysis roles in Massachusetts, New Hampshire, and Rhode Island (hybrid or remote).

CORE SKILLS

GIS PLATFORMS

ArcGIS Pro, ArcGIS Online, QGIS, PostGIS/PostgreSQL spatial databases

SCRIPTING & ANALYSIS

Python (GeoPandas, arcpy), R (sf, terra, ggplot2, Shiny, targets), SQL, reproducible ETL pipelines

REMOTE SENSING & FIELD

LiDAR terrain analysis, Landsat change detection, UAV imagery, CAD/GIS integration, Trimble GPS ground verification

WEB MAPPING

MapLibre GL JS, Leaflet, deck.gl, PMTiles vector tiles, HTML/CSS/JavaScript, real-time API integration

SPATIAL ANALYSIS

Geocoding at scale, network/routing analysis (OSRM), suitability modeling, hotspot & density analysis, random forest modeling, statistical correlation

DATA SOURCES

Census/ACS, MassGIS, USGS, OpenStreetMap, municipal ArcGIS services, live transit/AIS/ADS-B feeds

SELECTED PROJECTS — live at maxwellhowegis.com/work

Quabbin Reservoir LiDAR Study

Reproducible R pipeline (one command regenerates everything) producing an interactive 1 m bare-earth LiDAR explorer of the four drowned Swift River Valley towns, plus 20 figures: terrain, watershed, census history, and cross-sections.

Open Concord, NH — Geospatial Data Platform

Full-stack R + PostGIS platform: a {targets} ETL pipeline ingesting 165 layers from municipal ArcGIS, Census, CDC, EPA, USGS, and OSM into PostGIS, served through a Shiny/MapLibre frontend with live SQL queries, geocoding, and export.

Boston in Motion

One live map of everything moving through Boston — subway, commuter rail, ~150 bus routes, ferries, aircraft, harbor AIS traffic, and Bluebikes — streamed directly into the browser with no backend.

Massachusetts Education Atlas

Statewide interactive atlas of 1,700 schools and 274 districts with 40+ joined education and demographic metrics, layer panel, and classification controls.

Salem Pantry — Food Access Analysis — community partner project

Analyzed 150,000+ rows of client data in R to map service coverage and identify underserved areas; random forest model located demographically similar populations statewide.

Chronic Absenteeism Capstone — Lynn Public Schools

Geocoded a district's student population, built distance-band models, ran the statistical analysis in R, and presented an honest null result to district stakeholders.

PROFESSIONAL EXPERIENCE

Independent GIS Developer & Analyst — maxwellhowegis.com

2023 – Present

- Ship production web-GIS applications end to end: data acquisition, PostGIS design, statistical analysis, and interactive front ends (TappyMaps, Boston in Motion, Open Concord, PocketTiles Studio).
- Author reproducible, documented workflows — major projects rebuild from source data with a single command.

Mathematics & Statistics Educator — Lynn English High School, Lynn, MA

2022 – Present

- Teach probability and statistics; translate quantitative analysis for non-technical audiences daily.
- Built the Lynn Data Dive dashboard integrating enrollment, outcomes, finance, and demographic data across the 26 MA Gateway Cities.

Instructional Facilitator — LEAP for Education, Salem, MA

2023 – Present

- Designed and led data- and mapping-based enrichment programming, including GIS field exercises.

EDUCATION & CREDENTIALS

M.S., Geo-Information Science — Salem State University (LiDAR, remote sensing, spatial databases, surveying)

May 2025

B.S., Homeland Security (Intelligence Analysis) — Daniel Webster College · Massachusetts Mathematics Teaching License

2014